

LIMITS



SYNOPSIS

Neighbourhood of a real number:

- Let 'a' be a real number and δ be a positive real number, then $(a-\delta, a+\delta)$ is called δ -neighbourhood of a. The interval $(a-\delta, a)$ is called the left δ -neighbourhood of 'a' and the interval $(a, a+\delta)$ is called the right δ -neighbourhood of a. The deleted δ -neighbourhood of 'a' is denoted by $(a-\delta, a) \cup (a, a+\delta)$.

Limit of a function:

- Let f be a function defined over a deleted neighbourhood of the real number 'a' and ' l ' be a real number. If to every positive number ϵ (however small) there exists a positive number ' δ ' such that $|f(x)-l| < \epsilon$ for all x such that $0 < |x-a| < \delta$, we say that $f(x)$ tends to ' l ' as x tends to a and we write it as $\lim_{x \rightarrow a} f(x) = l$.

Right handed limit:

- Let $f(x)$ be a function defined in the interval $(a, a+h)$ and ' l ' be a real number. If to every positive number ϵ (however small), there corresponds a positive number δ such that $|f(x)-l| < \epsilon, \forall x \in (a, a+\delta)$ then we say that $f(x)$ tends to ' l ' as x approaches a through values higher than a and we denote

$$\lim_{x \rightarrow a^+} f(x) = l \quad (\text{or}) \quad \lim_{h \rightarrow 0} f(a+h) = l.$$

Left handed limit :

- Let $f(x)$ be a function defined in the interval $(a-h, a)$ and ' l ' be a real number. If to every positive number ϵ (however small), there

corresponds a positive number δ such that $|f(x)-l| < \epsilon$ for all x such that $x \in (a-\delta, a)$

we say that $f(x)$ tends to ' l ' as x approaches a through values less than a and we denote it as

$$\lim_{x \rightarrow a^-} f(x) = l \quad (\text{or}) \quad \lim_{h \rightarrow 0} f(a-h) = l$$

- Let $a, l \in R$ and f be a function defined on a deleted neighbourhood of 'a' then
- $$\lim_{x \rightarrow a} f(x) = l \Leftrightarrow \lim_{x \rightarrow a^+} f(x) = l = \lim_{x \rightarrow a^-} f(x)$$

- If $\lim_{x \rightarrow a} f(x)$ exists, then

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow 0} f(a+x) = \lim_{x \rightarrow 0} f(a-x)$$

Infinite limits :

- Let f be a function defined over a deleted neighbourhood of a . If for every positive number k (however large) there corresponds a positive number δ , such that $f(x) > k, \forall x$ such that $0 < |x-a| < \delta$, we say that $f(x) \rightarrow +\infty$ as $x \rightarrow a$. We write it as $\lim_{x \rightarrow a} f(x) = +\infty$

Similarly we define (i) $\lim_{x \rightarrow a} f(x) = -\infty$

$$(ii) \lim_{x \rightarrow +\infty} f(x) = +\infty, \quad \lim_{x \rightarrow +\infty} f(x) = -\infty$$

$$(iii) \lim_{x \rightarrow -\infty} f(x) = +\infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Limit of a function f as $x \rightarrow +\infty$ or $-\infty$:

- Let f be a function and l be a real number. If for every positive number ϵ there corresponds a positive number k (however large) such that $|f(x)-l| < \epsilon, \forall x > k$, then we say that $f(x)$ tends to l as x tends to ∞ . We write it as

$$\lim_{x \rightarrow +\infty} f(x) = l.$$

Similarly we define $\lim_{x \rightarrow -\infty} f(x) = l$.

Fundamental Theorem on Limits:

➤ If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then,

i) $\lim_{x \rightarrow a} \{f(x) + g(x)\} = l + m$

ii) $\lim_{x \rightarrow a} \{f(x) - g(x)\} = l - m$

iii) $\lim_{x \rightarrow a} \{f(x) \cdot g(x)\} = l \cdot m$

iv) $\lim_{x \rightarrow a} \{k \cdot f(x)\} = k \cdot \lim_{x \rightarrow a} f(x)$

v) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m}$, if $m \neq 0$

vi) $\lim_{x \rightarrow a} (f(x))^k = l^k$, if $k \in \mathbb{Q}$ and $l^k \in \mathbb{R}$

vii) $\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(m)$

viii) $\lim_{x \rightarrow a} (f(x))^{g(x)} = l^m$, if $l^m \in \mathbb{R}$

ix) If $f(x) \leq g(x)$ on a deleted nbd of 'a'

then $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$

x) $\lim_{x \rightarrow a} f(x) = l \Rightarrow \lim_{x \rightarrow a} |f(x)| = |l|$.

However the converse need not be true

Ex. $\lim_{x \rightarrow 0} \frac{1}{|x|} = \infty$ where as $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist.

Indeterminate forms :

➤ $\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 0^0, \infty^0$ and 1^∞ are called indeterminate forms

Standard Limits :

➤ For all real values of n, $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1}$

(Provided $n \cdot a^{n-1}$ is defined)

➤ $\lim_{x \rightarrow a} \frac{x^m - a^m}{x^n - a^n} = \frac{m}{n} a^{m-n}$, ($m > n$)

➤ If $0 < |x| < \frac{\pi}{2}$ and x is measured in radians.

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$,

$\lim_{x \rightarrow 0} \frac{\sin ax}{x} = a$ and $\lim_{x \rightarrow 0} \frac{\tan ax}{x} = a$

$\lim_{x \rightarrow 0} \frac{\sin x^0}{x} = \frac{\pi}{180}$ and $\lim_{x \rightarrow 0} \frac{\tan x^0}{x} = \frac{\pi}{180}$

➤ $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$ and $\lim_{x \rightarrow \infty} \frac{\cos x}{x} = 0$

➤ $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1$

➤ $\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = 1$

➤ $\lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log_e a$, ($a > 0$)

➤ $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x} = \log_e \left(\frac{a}{b} \right)$

➤ $\lim_{x \rightarrow 0} \frac{a^x - 1}{b^x - 1} = \log_b a$

➤ $\lim_{x \rightarrow a} \frac{|x - a|}{x - a}$ does not exist

➤ $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$, $\lim_{x \rightarrow 0} (1+ax)^{\frac{1}{x}} = e^a$

➤ $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$, $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} \right)^x = e^a$

➤ $f(x), g(x)$ are two polynomials such that degree of $f(x)$ is m and degree of $g(x)$ is n then

i) $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$ for $m < n$

ii) $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty$ for $m > n$ and coef of $x^m > 0$

iii) $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = -\infty$ for $m > n$ and coef of $x^m < 0$

iv) $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\text{coef of } x^m \text{ in } Nr}{\text{coef of } x^n \text{ in } Dr}$ for $m = n$

➤ $\lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = \infty$, $\lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0$

- $\lim_{x \rightarrow 0^+} e^{-\frac{1}{x}} = 0, \lim_{x \rightarrow 0^-} e^{-\frac{1}{x}} = \infty$
- $\lim_{n \rightarrow \infty} x^n = 0$ if $|x| < 1$
- $\lim_{n \rightarrow \infty} x^n = \infty$ if $|x| > 1$

W.E-1 : $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to

Sol: Given limit =

$$\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n} = \lim_{n \rightarrow \infty} 5 \left(1 + \left(\frac{4}{5} \right)^n \right)^{1/n} = 5$$

$$\left(\because \left(\frac{4}{5} \right)^n \rightarrow 0 \text{ as } n \rightarrow \infty \right)$$

W.E-2 : $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}}$ is equal to

Sol: $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}} = \lim_{n \rightarrow \infty} \frac{5 \cdot 5^n + 3^n - 4^n}{5^n + 2^n + 27 \cdot 9^n}$

$$= \lim_{n \rightarrow \infty} \frac{5 \cdot \frac{5^n}{9^n} + \frac{3^n}{9^n} - \frac{4^n}{9^n}}{\frac{5^n}{9^n} + \frac{2^n}{9^n} + 27} = \frac{0+0-0}{0+0+27} = 0$$

- $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right) = \text{Does not exist}$
- $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$

L'Hospital's Rule:

- If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

- If $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)}$$

- If $\lim_{x \rightarrow a} \frac{f^{(11)}(x)}{g^{(11)}(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

This can be continued till we finally arrive at a determinate result.

W.E-3 : Let $f(x)$ be a twice differentiable function and $f''(0) = 5$, then

$$\lim_{x \rightarrow 0} \frac{3f(x) - 4f(3x) + f(9x)}{x^2} \text{ is equal to}$$

Sol: $\lim_{x \rightarrow 0} \frac{3f(x) - 4f(3x) + f(9x)}{x^2} \left(\frac{0}{0} \text{ form} \right)$

$$= \lim_{x \rightarrow 0} \frac{3f'(x) - 12f'(3x) + 9f'(9x)}{2x} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{x \rightarrow 0} \frac{3f''(x) - 36f''(3x) + 81f''(9x)}{2}$$

$$= \frac{3f''(0) - 36f''(0) + 81f''(0)}{2}$$

$$= 24f''(0) = 24(5) = 120$$

Some useful results:

- Let $S = \{x, \sin x, \tan x, \sinh x, \tanh x, \sin^{-1} x, \tan^{-1} x, \sinh^{-1} x, \tanh^{-1} x\}$

If $f(x), g(x) \in S$ then $\lim_{x \rightarrow 0} \frac{f(mx)}{g(nx)} = \frac{m}{n}$, .

If $f_1(x), f_2(x), g_1(x), g_2(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{f_1(mx) \pm f_2(nx)}{g_1(px) \pm g_2(qx)} = \frac{m \pm n}{p \pm q}$$

- $\lim_{x \rightarrow 0} \frac{\sin ax}{\tan bx} = \frac{a}{b}, \lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2} = \frac{a^2}{2}$

W.E-4 : $\lim_{x \rightarrow 0} \frac{\sin 7x + \sin 5x}{\tan 5x - \tan 2x} = \frac{7+5}{5-2} = 4$

- If $f_1(x), f_2(x), g_1(x), g_2(x) \in S$ and $m + n = p + q$ then

$$\lim_{x \rightarrow 0} \frac{f_1^m(ax) f_2^n(bx)}{g_1^p(cx) g_2^q(dx)} = \frac{a^m b^n}{c^p d^q}$$

W.E-5 : $\lim_{x \rightarrow 0} \frac{\sin^3 2x \tan^2 3x}{x \sin^4 4x} = \frac{2^3 \times 3^2}{4^4} = \frac{9}{32}$

- If $g_1(x), g_2(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{1 - \cos ax}{g_1(cx) g_2(dx)} = \frac{a^2}{2cd}$$

$$\text{W.E-6: } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin 3x} = \frac{1}{6}$$

➤ If $g_1(x), g_2(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{1 - \cos^n(ax)}{g_1(cx)g_2(dx)} = \frac{na^2}{2cd}$$

$$\text{W.E-7: } \lim_{x \rightarrow 0} \frac{1 - \cos^3 2x}{\sin 5x \tan 7x} = \frac{3 \times 2^2}{2 \times 5 \times 7} = \frac{6}{35}$$

➤ If $g_1(x), g_2(x), \dots, g_{2n}(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{1 - \cos(ax^n)}{g_1(c_1x) \cdot g_2(c_2x) \cdots g_{2n}(c_{2n}x)} = \frac{a^2}{2c_1c_2 \cdots c_{2n}}$$

$$\text{W.E-8: } \lim_{x \rightarrow 0} \frac{1 - \cos(2x^3)}{x \sin^2(2x) \tan^3(3x)} = \frac{4}{2 \times 2^2 \times 3^3} = \frac{1}{54}$$

➤ If $g_1(x), g_2(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{f(cx)g(dx)} = \frac{b^2 - a^2}{2cd}$$

$$\text{W.E-9: } \lim_{x \rightarrow 0} \frac{\cos 3x - \cos 5x}{x^2} = \frac{25 - 9}{2} = 8$$

➤ If $g_1(x), g_2(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{\cos^n ax - \cos^n bx}{g_1(cx) \cdot g_2(dx)} = \frac{n(b^2 - a^2)}{2cd}$$

$$\text{W.E-10: } \lim_{x \rightarrow 0} \frac{\cos^3 3x - \cos^3 5x}{x^2} = \frac{3(25 - 9)}{2} = 24$$

➤ If $g_1(x), g_2(x), \dots, g_{2n}(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{\cos(ax^n) - \cos(bx^n)}{g_1(c_1x) \cdot g_2(c_2x) \cdots g_{2n}(c_{2n}x)} = \frac{b^2 - a^2}{2c_1c_2 \cdots c_{2n}}$$

$$\text{W.E-11: } \lim_{x \rightarrow 0} \frac{\cos(2x^3) - \cos(5x^3)}{x \sin^2(2x) \tan^3(3x)} = \frac{25 - 4}{2 \times 2^2 \times 3^3} = \frac{7}{18 \times 4} = \frac{7}{72}$$

➤ If $g_1(x) \in S$ then

$$\lim_{x \rightarrow 0} \frac{\tan^n(ax) - \sin^n(ax)}{[g(x)]^{n+2}} = \frac{na^{n+2}}{2}$$

$$\text{W.E-12: } \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x^n} - \sqrt{1-x^n}}{x^n} = 1$$

$$\text{W.E-13: } \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{x^2} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sqrt[n]{a+x^m} - \sqrt[n]{a-x^m}}{x^m} = \frac{2}{n} a^{\frac{1}{n}-1}$$

$$\text{W.E-14: } \lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{x} = a^{\frac{1}{2}-1} = \frac{1}{\sqrt{a}}$$

$$\text{If } g(x) \in S \text{ then } \lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{g(x)} = \frac{1}{\sqrt{a}}$$

$$\text{W.E-15: } \lim_{x \rightarrow 0} \frac{\sqrt{3+x} - \sqrt{3-x}}{\sin x} = \frac{1}{\sqrt{3}}$$

$$\text{If } g(x) \in S \text{ then } \lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{g(x)} = \frac{1}{2\sqrt{a}}$$

$$\text{W.E-16: } \lim_{x \rightarrow 0} \frac{\sqrt{2+x} - \sqrt{2}}{x} = \frac{1}{2\sqrt{2}}$$

$$\lim_{x \rightarrow 0} \frac{x \cdot a^{\alpha x} - x}{1 - \cos(mx)} = \frac{2\alpha}{m^2} \log a$$

$$\text{W.E-17: } \lim_{x \rightarrow 0} \frac{x \cdot 2^{3x} - x}{1 - \cos(3x)} = \frac{2 \times 3}{3^2} \log 2 = \frac{2}{3} \log 2$$

➤ $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \infty$ then

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} g(x)[f(x)-1]}$$

W.E-18: If $\lim_{x \rightarrow 0} (1 + ax + bx^2)^{2/x} = e^3$, then the values of a and b are

Sol: Let $\lim_{x \rightarrow 0} (1 + ax + bx^2)^{2/x}$ is of the form 1^∞

$$e^{\lim_{x \rightarrow 0} (1 + ax + bx^2 - 1) \cdot \frac{2}{x}} = e^{\lim_{x \rightarrow 0} (2a + 2bx)} = e^{2a} = e^3 \text{ (given)} \quad \therefore a = 3/2 \text{ and } b \in \mathbb{R}$$

➤ $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, then

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} g(x) \log f(x)} \quad (f(x) > 0)$$

W.E-19: $\lim_{x \rightarrow 1} (1-x^2)^{\frac{1}{\log(1-x)}} =$

Sol :
$$\lim_{x \rightarrow 1} \frac{\log(1-x^2)}{\log(1-x)} = \lim_{x \rightarrow 1} \left[\frac{\log(1-x) + \log(1+x)}{\log(1-x)} \right]$$

$$= e^{\lim_{x \rightarrow 1} \left[1 + \frac{\log(1+x)}{\log(1-x)} \right]} = e^{1+0} = e$$

➤
$$\lim_{x \rightarrow 0} \left[\frac{a_1^x + a_2^x + a_3^x + \dots + a_n^x}{n} \right]^{\frac{1}{x}} = (a_1 \cdot a_2 \dots a_n)^{\frac{1}{n}}$$

➤
$$\lim_{x \rightarrow \infty} \left[\frac{\frac{1}{a_1^x} + \frac{1}{a_2^x} + \frac{1}{a_3^x} + \dots + \frac{1}{a_n^x}}{n} \right]^x = (a_1 \cdot a_2 \dots a_n)^{\frac{1}{n}}$$

W.E-20 : Evaluate $\lim_{x \rightarrow 0} \left(\frac{2^x + 2^{2x} + 2^{3x}}{3} \right)^{1/x}$

Sol:
$$\lim_{x \rightarrow 0} \left(\frac{2^x + 2^{2x} + 2^{3x}}{3} \right)^{1/x} = (2 \cdot 4 \cdot 8)^{1/3} = 4$$

W.E-21 : $\lim_{x \rightarrow 0} [\cos x + m \sin ax]^{\frac{n}{x}} = e^{amn}$

➤
$$\lim_{x \rightarrow \infty} a^x = \begin{cases} 0, & 0 < a < 1 \\ 1, & a = 1 \\ \infty, & a > 1 \end{cases}$$

 does not exist, $a < 0$

Some frequently used expansions:

➤ i) $(1+x)^p =$

$$1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \dots \infty,$$

if $|x| < 1$

ii) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$

iii) $a^x = 1 + \frac{x}{1!} \cdot \log_e a + \frac{x^2}{2!} (\log_e a)^2 + \dots \infty$

iv) $\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty$

v) $\log_e(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty\right)$

vi) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \infty$

vii) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$

viii) $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots \infty$

ix) $\sin^{-1} x = x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{x^5}{5} + \dots \infty$

x) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \infty$

W.E-22: Evaluate $\lim_{x \rightarrow 0} \left(\frac{(1+x)^{\frac{1}{x}} - e + \frac{ex}{2}}{\sin^2 x} \right)$

Sol :
$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{x^2} \cdot \left(\frac{x^2}{\sin^2 x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{\sin^2 x} \cdot 1$$

$$= \lim_{x \rightarrow 0} \frac{e \left[1 - \frac{x}{2} + \frac{11}{24}x^2 \dots \right] - e + \frac{ex}{2}}{x^2} = \frac{11e}{24}$$

W.E-23: Find $\lim_{x \rightarrow 0} \left(\frac{\sin x - x + \frac{x^3}{6}}{x^5} \right)$

Sol :
$$\lim_{x \rightarrow 0} \frac{\left(\sin x - x + \frac{x^3}{6} \right)}{x^5}$$

$$= \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right) - x + \frac{x^3}{6}}{x^5}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{\left(\frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)}{x^5} \\
 &= \lim_{x \rightarrow 0} \left(\frac{1}{5!} - \frac{x^2}{7!} + \text{terms containing positive integral powers of } x \right) \\
 &= \frac{1}{5!} = \frac{1}{120}
 \end{aligned}$$

W.E-24: The value of $\lim_{n \rightarrow \infty} \left[\sqrt[3]{(n+1)^2} - \sqrt[3]{(n-1)^2} \right]$ is

Sol:

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \left[\sqrt[3]{(n+1)^2} - \sqrt[3]{(n-1)^2} \right] \\
 &= \lim_{n \rightarrow \infty} n^{2/3} \left[\left(1 + \frac{1}{n} \right)^{2/3} - \left(1 - \frac{1}{n} \right)^{2/3} \right] \\
 &= \lim_{n \rightarrow \infty} n^{2/3} \left[\left(1 + \frac{2}{3} \cdot \frac{1}{n} + \frac{2 \left(\frac{2}{3} - 1 \right)}{2!} \frac{1}{n^2} + \dots \right) \right. \\
 &\quad \left. - \left(1 - \frac{2}{3} \cdot \frac{1}{n} + \frac{2 \left(\frac{2}{3} - 1 \right)}{2!} \frac{1}{n^2} + \dots \right) \right] \\
 &= \lim_{n \rightarrow \infty} n^{2/3} \left[\frac{4}{3} \cdot \frac{1}{n} + \frac{8}{81} \cdot \frac{1}{n^3} + \dots \right] \\
 &= \lim_{n \rightarrow \infty} \left[\frac{4}{3} \cdot \frac{1}{n^{1/3}} + \frac{8}{81} \cdot \frac{1}{n^{7/3}} + \dots \right] = 0
 \end{aligned}$$

Sandwich theorem or Squeezeprinciple:

➤ If f, g, h are functions such that

$$f(x) \leq g(x) \leq h(x)$$

then $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x) \leq \lim_{x \rightarrow a} h(x)$ and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = l \text{ then } \lim_{x \rightarrow a} g(x) = l$$

W.E-25: If $3 - \frac{x^2}{12} \leq f(x) \leq 3 + \frac{x^3}{9}$ for all $x \neq 0$,

then the value of $\lim_{x \rightarrow 0} f(x)$ is

Sol: According to the equation,

$$\lim_{x \rightarrow 0} \left(3 - \frac{x^2}{12} \right) \leq \lim_{x \rightarrow 0} f(x) \leq \lim_{x \rightarrow 0} \left(3 + \frac{x^3}{9} \right)$$

$$\Rightarrow (3 - 0) \leq \lim_{x \rightarrow 0} f(x) \leq (3 + 0)$$

$$\text{Hence, } \lim_{x \rightarrow 0} f(x) = 3$$

➤ $\lim_{n \rightarrow \infty} \frac{[1^k x] + [2^k x] + \dots + [n^k x]}{n^{k+1}} = \frac{x}{k+1} \quad (k \in \mathbb{N})$

W.E-26: Show that $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2} = \frac{x}{2}$.

Sol : For $r=1, 2, 3, \dots, n, rx - 1 < [rx] \leq rx$

$$\Rightarrow \sum_{r=1}^n (rx - 1) < \sum_{r=1}^n [rx] \leq \sum_{r=1}^n (rx)$$

$$\Rightarrow \frac{\frac{n(n+1)x}{2} - n}{n^2} < \frac{\sum_{r=1}^n [rx]}{n^2} \leq \frac{\frac{n(n+1)x}{2}}{2n^2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\left(1 + \frac{1}{n} \right) \frac{x}{2} - \frac{1}{n} \right) \leq \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n [rx]}{n^2} \leq \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \frac{x}{2}$$

(Note that x is a constant and n is a variable)

$$\Rightarrow \frac{x}{2} \leq \lim_{n \rightarrow \infty} \frac{[1.x] + [2.3] + \dots + [n.x]}{n^2} \leq \frac{x}{2}$$

∴ By sandwich theorem,

$$\lim_{n \rightarrow \infty} \frac{[1x] + [2x] + \dots + [nx]}{n^2} = \frac{x}{2}$$



LEVEL - I (C.W)

EVALUATION OF ALGEBRA OF LIMITS

1. $\lim_{x \rightarrow 0} \frac{\sqrt[k]{1+x} - 1}{x}$ (K is a positive integer)

- 1) K 2) $-K$ 3) $\frac{1}{K}$ 4) $-\frac{1}{K}$

2. $\lim_{x \rightarrow 1} \frac{(2x-3)(\sqrt{x}-1)}{2x^2+x-3} =$
 1) $\frac{1}{10}$ 2) $-\frac{1}{10}$ 3) $\frac{2}{5}$ 4) $-\frac{2}{5}$
3. $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+\sin x} - \sqrt[3]{1-\sin x}}{x} =$
 1) 0 2) 1 3) $\frac{2}{3}$ 4) $\frac{3}{2}$
4. If $\lim_{x \rightarrow 5} \frac{x^k - 5^k}{x-5} = 500$, then the positive integral value of k is
 1) 3 2) 4 3) 5 4) 6
5. $\lim_{x \rightarrow 1} \frac{\sqrt{x^2-1} + \sqrt{x-1}}{\sqrt{x^2-1}} =$
 1) $1 + \frac{1}{\sqrt{2}}$ 2) $1 - \frac{1}{\sqrt{2}}$ 3) $-1 + \frac{1}{\sqrt{2}}$ 4) $-1 - \frac{1}{\sqrt{2}}$
6. If $a > 0$ and $\lim_{x \rightarrow a} \frac{a^x - x^a}{x^x - a^a} = -1$ then $a =$
 1) 0 2) 1 3) e 4) $2e$
7. $\lim_{x \rightarrow 0} \frac{x(1 - \sqrt{1-x^2})}{\sqrt{1-x^2}(\sin^{-1}(x)^3)} =$
 1) 1 2) $\frac{1}{2}$ 3) $-\frac{1}{2}$ 4) -1
8. If $\lim_{x \rightarrow 0} \left(\frac{\cos 4x + a \cos 2x + b}{x^4} \right)$ is finite then the value of a, b respectively
 1) 5 2) -5, -4 3) -4, 3 4) 4, 5
9. $\lim_{x \rightarrow 1} \left(\sec\left(\frac{\pi x}{2}\right) \log x \right)$ is
 1) $-\frac{2}{\pi}$ 2) $-\frac{\pi}{2}$ 3) $\frac{2}{\pi}$ 4) $\frac{\pi}{2}$
10. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \sqrt{1-x+x^2}}{3^x - 1} =$ (Eamcet2014)
 1) $\log 9$ 2) $\frac{1}{\log 9}$ 3) $\log 3$ 4) $\frac{1}{\log 3}$

EVALUATION OF LEFT & RIGHT HAND LIMITS

11. $\lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x^2}} =$
 1) 1 2) -1 3) 0 4) doesn't exist
12. $\lim_{x \rightarrow 2^+} \left(\frac{[x]^3}{3} - \left[\frac{x}{3} \right]^3 \right)$ is (where $[]$ is g.i.f)
 1) 0 2) $\frac{64}{27}$ 3) $\frac{8}{3}$ 4) $\frac{10}{3}$
13. If $f: R \rightarrow R$ is defined by (where $[]$ is g.i.f) $f(x) = [x-3] + |x-4|$ for $x \in R$ then $\lim_{x \rightarrow 3^-} f(x) =$ [EAM- 2008]
 1) -2 2) -1 3) 0 4) 2.

EVALUATION OF TRIGONOMETRIC LIMITS

14. $\lim_{x \rightarrow 0} \frac{e^x - e^{\sin x}}{2(x - \sin x)} =$ [EAMCET 2007]
 1) -1/2 2) 1/2 3) 1 4) 3/2
15. $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x} =$
 1) 1/2 2) 3/2 3) 3/4 4) 1/4
16. $\lim_{x \rightarrow 1} (1-x) \tan\left(\frac{\pi x}{2}\right) =$
 1) π 2) 2π 3) $\frac{\pi}{2}$ 4) $\frac{2}{\pi}$
17. $\lim_{x \rightarrow \frac{\pi}{6}} \frac{3 \sin x - \sqrt{3} \cos x}{6x - \pi} =$
 1) $\sqrt{3}$ 2) $\frac{1}{\sqrt{3}}$ 3) $-\sqrt{3}$ 4) $-\frac{1}{\sqrt{3}}$
18. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x - \cos x}{\left(\frac{\pi}{2} - x\right)^3} =$
 1) $-\frac{1}{2}$ 2) $\frac{1}{2}$ 3) 2 4) -2
19. $\lim_{x \rightarrow 0} \frac{\sec 4x - \sec 2x}{\sec 3x - \sec x} =$
 1) $\frac{3}{2}$ 2) $\frac{2}{3}$ 3) $\frac{1}{3}$ 4) $\frac{3}{4}$

20. $\lim_{x \rightarrow 0} \frac{3 \sin(x^g) - \sin(3x^g)}{x^3} =$

- 1) $\left(\frac{\pi}{200}\right)^3$ 2) $4\left(\frac{\pi}{200}\right)^3$ 3) $\frac{\pi}{200}$ 4) $\frac{\pi}{100}$

21. $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2} =$ (MAINS-2014)

- 1) $-\pi$ 2) π 3) $\frac{\pi}{2}$ 4) 1

EVALUATION OF EXPONENTIAL & LOGARITHMIC LIMITS

22. $\lim_{x \rightarrow 0} \log \left| \frac{\log(1+x)}{x} \right| =$

- 1) 0 2) 1 3) e 4) $1/e$

23. The value of $\lim_{x \rightarrow 2} \frac{2^x + 2^{3-x} - 6}{\sqrt{2^{-x}} - 2^{1-x}}$ is

- 1) 16 2) 8 3) 4 4) 2

24. $\lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{\log(1+x)} =$

- 1) 1 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) 2

25. $\lim_{x \rightarrow 0} \frac{10^x - 2^x - 5^x + 1}{x \tan x}$ is

- 1) $\log 2$ 2) $\frac{\log 2}{\log 5}$
3) $(\log 2)(\log 5)$ 4) $\log 10$

LIMITS OF THE FORM TENDS TO INFINITE

26. $\lim_{n \rightarrow \infty} \left(\frac{1}{a(a+d)} + \frac{1}{(a+d)(a+2d)} + \dots \text{to } n \text{ terms} \right) (a > 0, d > 0) =$

- 1) $\frac{1}{a}$ 2) $\frac{1}{d}$ 3) $\frac{a}{d}$ 4) $\frac{1}{ad}$

27. $\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}}{1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n}} =$

- 1) $4/3$ 2) $3/4$ 3) $1/2$ 4) 0

28. $\lim_{x \rightarrow \infty} \left(\frac{x^2 \sin\left(\frac{1}{x}\right) - x}{1 - |x|} \right) =$

- 1) 0 2) 1 3) -1 4) 2

29. If $0 < h < q$ then $\lim_{n \rightarrow \infty} (q^n + h^n)^{\frac{1}{n}} =$

- 1) e 2) h 3) q 4) 0

30. $\lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x}) =$

- 1) 2 2) -2 3) 0 4) 1

31. $\lim_{x \rightarrow \infty} x \cos\left(\frac{\pi}{8x}\right) \sin\left(\frac{\pi}{8x}\right) =$

- 1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{8}$ 4) $\frac{\pi}{4}$

32. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + ax + a^2} - \sqrt{x^2 + a^2}) =$

- 1) 0 2) $\frac{a}{2}$ 3) $-\frac{a}{2}$ 4) a

33. $\lim_{x \rightarrow \infty} \frac{x - \log x}{x + \log x} =$

- 1) 1 2) -1 3) 0 4) 2

34. For $a > 1$ then $\lim_{x \rightarrow \infty} \frac{a^x - a^{-x}}{a^x + a^{-x}} =$

- 1) 1 2) $1/2$ 3) $1/3$ 4) $1/15$

35. $\lim_{x \rightarrow \infty} \frac{2x + 7 \sin x}{4x + 3 \cos x} =$

- 1) 1 2) -1 3) $1/2$ 4) $-1/2$

36. $\lim_{x \rightarrow \infty} \frac{8|x| + 3x}{3|x| - 2x} =$

- 1) 11 2) 8 3) 0 4) $\frac{1}{8}$

37. $\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}} =$

- 1) 10 2) 100 3) 1000 4) 1

38. $\lim_{n \rightarrow \infty} \frac{3 \cdot 2^{n+1} - 4 \cdot 5^{n+1}}{5 \cdot 2^n + 7 \cdot 5^n} =$

- 1) $-20/7$ 2) $20/7$ 3) $10/7$ 4) $-10/7$

39. $\lim_{n \rightarrow \infty} \frac{1}{n^3} \left\{ 1 + 3 + 6 + \dots + \frac{n(n+1)}{2} \right\} =$

- 1) 0 2) 2 3) $\frac{1}{6}$ 4) $\frac{1}{3}$

LIMIT OF THE FORM $0^0, 1^\infty, \infty^0$

40. $\lim_{x \rightarrow 1} (2-x)^{\tan\left(\frac{\pi x}{2}\right)} =$
 1) $\frac{1}{e^\pi}$ 2) $e^{\frac{2}{\pi}}$ 3) $-e^{\frac{2}{\pi}}$ 4) e
41. $\lim_{x \rightarrow 0^+} (\sin x)^{\tan x} =$
 1) e 2) e^2 3) -1 4) 1
42. $\lim_{n \rightarrow \infty} (\pi n)^{2/n} =$
 1) 0 2) 1 3) 2 4) 3
43. $\lim_{x \rightarrow \infty} \left(\frac{a^{1/x} + b^{1/x} + c^{1/x}}{3} \right)^x =$ (where a, b, c are real and non-zero) [EAMCET 1998]
 1) 0 2) $(abc)^{1/3}$ 3) $(abc)^{-1/3}$ 4) 1
44. $\lim_{n \rightarrow \infty} \left(1 + \sin\left(\frac{a}{n}\right) \right)^n =$
 1) e 2) e^a 3) a^e 4) a

L.H.RULE

45. If $f(9) = 9$, $f'(9) = 4$, $\lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3} =$
 1) 4 2) $\frac{1}{4}$ 3) $\frac{1}{2}$ 4) $-\frac{1}{2}$
46. If $f(a) = 2$, $f'(a) = 1$, $g(a) = -1$, $g'(a) = 2$ then
 $\lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x - a} =$
 1) $\frac{1}{5}$ 2) 5 3) $-\frac{1}{5}$ 4) -5
47. If $f(x) = x \tan^{-1} x$ then $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} =$
 (Eamcet-2014)
 1) $\frac{\pi}{4}$ 2) $\frac{\pi+1}{4}$ 3) $\frac{\pi+2}{4}$ 4) $\frac{\pi+3}{4}$

PROBLEMS USING EXPANSIONS

48. $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{\sin^2 2x} =$
 1) $\frac{3}{2}$ 2) $\frac{5}{2}$ 3) $\frac{7}{4}$ 4) $\frac{9}{2}$

PROBLEMS USING SANDWICH THEOREM

49. $\lim_{x \rightarrow 0} x^3 \cos \frac{2}{x} =$
 1) 0 2) 1 3) ∞ 4) does not exist
50. $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+2} + \frac{n}{n^2+3} + \dots + \frac{n}{n^2+n} \right) =$
 1) 0 2) 1 3) ∞ 4) does not exist

LEVEL-I (C.W)-KEY

- 01) 3 02) 2 03) 3 04) 2 05) 1 06) 2
 07) 2 08) 3 09) 1 10) 2 11) 4 12) 3
 13) 3 14) 2 15) 3 16) 4 17) 2 18) 2
 19) 1 20) 2 21) 2 22) 1 23) 2 24) 4
 25) 3 26) 4 27) 1 28) 1 29) 3 30) 3
 31) 3 32) 2 33) 1 34) 1 35) 3 36) 1
 37) 2 38) 1 39) 3 40) 2 41) 4 42) 2
 43) 2 44) 2 45) 1 46) 2 47) 3 48) 3
 49) 1 50) 2

LEVEL-I (C.W)-HINTS

1. Using L-Hospital rule
 2. Given limit

3. Using L-Hospital rule

4. Using

- 5.

6. Use L-Hospital rule

7. On rationalising given limit

8. By L-Hospital rule

LIMITS

9.

10. On rationalising

11. does not exist

12. For

13. Definition of step function

14.

15. Given limit is

16.

17. By L-Hospital rule

18.

19.

20.

10

JEE-MAIN-JR-MATHS VOL-IV

21.

22.

23.

[Multiplying and by]

$$24. \lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{\log(1+x)} = \lim_{x \rightarrow 0} \frac{e^x + \cos x}{\frac{1}{1+x}} = 2$$

25.

26.

27. Given limit (if $r < 1$ then sum of terms in G.P

is

=

28. Divide with

29. Taking common

30. Using the formula $\sin C - \sin D$ and $\sin x$ is bounded

31.

32. Rationalising

33. Use L-Hospital rule

34. Divide with $\frac{1}{x}$ 35. Divide with $\frac{1}{x}$

$$36. \lim_{x \rightarrow \infty} \frac{8|x| + 3x}{3|x| - 2x} = \lim_{x \rightarrow \infty} \frac{8x + 3x}{3x - 2x} = 11$$

37. Divide with $\frac{1}{x}$ 38. Taking $\frac{1}{x}$ common and simplify39. Given limit is $\frac{1}{x}$

40.

and using L hospital rule.

41.

42. apply log on both sides

43.

44.

45.

46.

By $\frac{1}{x}$ 47. Find $\frac{1}{x}$ use L.H. Rule

48.

49.

for $\frac{1}{x}$ and $\frac{1}{x}$ for $\frac{1}{x}$.where $\frac{1}{x}$ (or) $\frac{1}{x}$ finite number

between -1 and +1 = 0

50. Use Sandwich Theorem

**LEVEL-I - (H.W)****EVALUATION OF ALGEBRA OF LIMITS**

$$1. \lim_{x \rightarrow a} \frac{x^{\frac{5}{8}} - a^{\frac{5}{8}}}{x^{\frac{1}{3}} - a^{\frac{1}{3}}} =$$

$$1) \frac{15}{8} a^{\frac{7}{24}} \quad 2) \frac{15}{4} a^{\frac{7}{24}} \quad 3) -\frac{15}{8} a^{\frac{7}{24}} \quad 4) \frac{15}{4} a^{-\frac{7}{24}}$$

$$2. \lim_{x \rightarrow 2} \frac{7x^2 - 11x - 6}{3x^2 - x - 10} =$$

$$1) \frac{17}{11} \quad 2) \frac{11}{17} \quad 3) \frac{17}{14} \quad 4) -\frac{17}{11}$$

$$3. \lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{m}} - (1-x)^{\frac{1}{m}}}{(1+x)^{\frac{1}{n}} - (1-x)^{\frac{1}{n}}} =$$

$$1) 1 \quad 2) \frac{m+n}{m-n} \quad 3) \frac{n}{m} \quad 4) \frac{m}{n}$$

$$4. \lim_{x \rightarrow 0} \frac{\sqrt{4+x} - \sqrt[3]{8+3x}}{x} =$$

$$1) -\frac{1}{2} \quad 2) \frac{1}{2} \quad 3) -3 \quad 4) 0$$

$$5. \lim_{x \rightarrow 0} \frac{\sqrt{1+\tan x} - \sqrt{1-\tan x}}{\sin x} =$$

$$1) 0 \quad 2) 1 \quad 3) 2 \quad 4) \frac{1}{2}$$

$$6. \lim_{x \rightarrow -1} \frac{x+1}{\sqrt{x^2+3}-2} =$$

$$1) -2 \quad 2) 1/2 \quad 3) 2 \quad 4) 0$$

7. If $7 - \frac{x^2}{12} \leq f(x) \leq 7 + \frac{x^3}{5}$ for all $x \neq 0$,

then $\lim_{x \rightarrow 0} f(x) =$

- 1) 3 2) 5 3) 7 4) 9

EVALUATION OF LEFT & RIGHT HAND LIMITS

8. If $f(x) = \begin{cases} \frac{x^2}{2}, & \text{if } 0 \leq x < 1 \\ 2x^2 - 2x + \frac{3}{2}, & \text{if } 1 \leq x \leq 2 \end{cases}$

then $\lim_{x \rightarrow 1} f(x) =$

- 1) $1/2$ 2) $3/2$ 3) does not exist 4) $-1/2$

9. $\lim_{x \rightarrow 0} \frac{\sin[\cos x]}{1 + [\cos x]}$ is (where $[]$ is g.i.f)

- 1) 1 2) 0 3) does not exist 4) 2

EVALUATION OF TRIGONOMETRIC LIMITS

10. $\lim_{x \rightarrow 0} \left(\frac{\sec ax - \sec bx}{x^2} \right) =$

- 1) $\frac{a^2 - b^2}{2}$ 2) $\frac{b^2 - a^2}{2}$ 3) 0 4) 1

11. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{\left(\frac{\pi}{4} - x \right) (\cos x + \sin x)} =$

- 1) 2 2) 1 3) 0 4) 3

12. $\lim_{x \rightarrow 0} \frac{\sin x \sin\left(\frac{\pi}{3} + x\right) \sin\left(\frac{\pi}{3} - x\right)}{x} =$

- 1) $\frac{3}{4}$ 2) $\frac{1}{4}$ 3) $\frac{4}{3}$ 4) 0

13. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x} =$

- 1) $10/3$ 2) $3/10$ 3) $6/5$ 4) 56

14. $\lim_{x \rightarrow 5} \frac{\sin^2(x-5) \tan(x-5)}{(x^2 - 25)(x-5)} =$

- 1) 1 2) $1/10$ 3) 0 4) -6

15. $\lim_{x \rightarrow 0} \frac{1 - \cos 2x^\circ}{x^2} =$

- 1) 0 2) 1 3) $\left(\frac{\pi}{180}\right)^2$ 4) $2\left(\frac{\pi}{180}\right)^2$

16. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \log(1+x)} =$

- 1) 1 2) 0 3) -1 4) $\frac{1}{2}$

17. $\lim_{x \rightarrow 0} \frac{1}{x^8} \left[1 - \cos\left(\frac{x^2}{2}\right) \right] \left[1 - \cos\left(\frac{x^2}{4}\right) \right] =$

- 1) $\frac{1}{8}$ 2) $\frac{1}{8^2}$ 3) $\frac{2}{8^3}$ 4) $\frac{2}{8^4}$

18. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \cot^3 x}{2 - \cot x - \cot^3 x} =$

- 1) $\frac{11}{4}$ 2) $\frac{3}{4}$ 3) $\frac{1}{2}$ 4) 0

EVALUATION OF EXPONENTIAL & LOGARITHMIC LIMITS

19. $\lim_{x \rightarrow 0} \frac{\log_e(1+x)}{3^x - 1} =$

- 1) $\log_e 3$ 2) 0 3) 1 4) $\log_3 e$

20. $\lim_{x \rightarrow 0} \frac{e^{\alpha x} - e^{\beta x}}{\sin \alpha x - \sin \beta x} =$

- 1) 0 2) $1/2$ 3) $1/3$ 4) 1

21. $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{3x} =$

- 1) $2/3$ 2) 6 3) $3/2$ 4) $1/6$

22. $\lim_{x \rightarrow 0} \frac{\log(1+ax) - \log(1+bx)}{x} =$

- 1) $a+b$ 2) $a-b$ 3) $-(a+b)$ 4) ab

23. $\lim_{x \rightarrow 0} \frac{4^x - 9^x}{x(4^x + 9^x)} =$

- 1) $\log\left(\frac{3}{2}\right)$ 2) $\log\left(\frac{2}{3}\right)$ 3) $\log\left(\frac{4}{3}\right)$ 4) $\log 2$

24. $\lim_{x \rightarrow 0} \frac{a^x - 1}{b^x - 1} =$

- 1) $\log_b a$ 2) $\log_a b$
 3) $\log_e ab$ 4) $\log_e \left(\frac{a}{b}\right)$

LIMITS OF THE FORM TENDS TO INFINITE

25. $\lim_{x \rightarrow \infty} 2x(\sqrt{x^2 + 1} - x) =$

- 1) 1 2) $1/2$ 3) 0 4) -1

26. $\lim_{x \rightarrow \infty} \frac{\sqrt[7]{x^7 - 1} + \sqrt[5]{x^5 + 2} + \sqrt[9]{x^9 - 2}}{\sqrt[6]{x^6 + 1} + \sqrt[5]{x^5 + 1} - \sqrt[4]{x^4 + 4}} =$

- 1) 3 2) -3 3) $1/3$ 4) $-1/3$

27. $\lim_{x \rightarrow \infty} 5^x \sin\left(\frac{a}{5^x}\right) =$

- 1) 1 2) a 3) ∞ 4) not defined

28. $\lim_{n \rightarrow \infty} \frac{2^{\frac{1}{n}} - 1}{\frac{1}{2^n} + 1} =$

- 1) 1 2) $\frac{1}{2}$ 3) -1 4) 0

29. $\lim_{n \rightarrow \infty} \frac{(1+2+3+\dots+n \text{ terms})(1^2+2^2+\dots+n \text{ terms})}{n(1^3+2^3+\dots+n \text{ terms})} =$

- 1) $\frac{3}{2}$ 2) $\frac{2}{3}$ 3) 1 4) 0

30. $\lim_{n \rightarrow \infty} \left(\frac{1}{1.3} + \frac{1}{3.5} + \dots + \frac{1}{(2n-1)(2n+1)} \right) =$

- 1) 1 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $\frac{1}{4}$

31. $\lim_{n \rightarrow \infty} \frac{1+3+5+\dots+(2n-1)}{2+4+6+\dots+2n} =$

- 1) 0 2) 1 3) -1 4) 5

32. $\lim_{n \rightarrow \infty} \frac{(n+2)! + (n+1)!}{(n+3)!} =$

- 1) 1 2) $1/2$ 3) 0 4) .2

LIMIT OF THE FORM $0^0, 1^\infty, \infty^0$

33. If $\lim_{x \rightarrow \infty} \left(1 + \frac{\lambda}{x} + \frac{\mu}{x^2} \right)^{2x} = e^2$ then

- 1) $\lambda = 1, \mu = 2$ 2) $\lambda = 2, \mu = 1$

3) $\lambda = 1, \mu =$ any real constant

4) $\lambda = \mu = 1$

34. $\lim_{x \rightarrow \infty} \left(\frac{x+a}{x+b} \right)^{x+b} =$

- 1) 1 2) e^{b-a} 3) e^{a-b} 4) e^b

35. $\lim_{x \rightarrow \pi} (1 - 4 \tan x)^{\cot x} =$

- 1) e 2) e^4 3) e^{-1} 4) e^{-4}

36. $\lim_{x \rightarrow \infty} \left(\frac{x^2+1}{x^2-1} \right)^{x^2} =$

- 1) e 2) $1/e$ 3) e^2 4) e^{-2}

37. $\lim_{x \rightarrow \infty} \left(\frac{3x-4}{3x+2} \right)^{\frac{x+1}{3}} =$

- 1) $e^{-2/3}$ 2) $e^{3/2}$ 3) $e^{2/3}$ 4) e

38. $\lim_{x \rightarrow 1} x^{\left(\frac{1}{1-x^2} \right)} =$

- 1) $e^{-\frac{1}{2}}$ 2) $e^{\frac{2}{3}}$ 3) $e^{\frac{3}{2}}$ 4) e

39. $\lim_{x \rightarrow 1} \left(\frac{1+x}{2+x} \right)^{\left(\frac{1-\sqrt{x}}{1-x} \right)} =$

- 1) $\frac{2}{3}$ 2) $\sqrt{\frac{2}{3}}$ 3) $\frac{2}{\sqrt{3}}$ 4) $\frac{\sqrt{2}}{3}$

L.H.RULE

40. If $f(x) = x^2 + x + 1$, then $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} =$

- 1) 3 2) 0 3) -1 4) 2

41. $f(x) = x^2$, $\phi(a) = b^2$, $f'(a) = n\phi'(a)$ and

$\Phi'(a) \neq 0$ then $\lim_{x \rightarrow a} \left(\frac{\sqrt{f(x)} - a}{\sqrt{\phi(x)} - b} \right) =$

- 1) $\frac{b}{a}$ 2) $\frac{nb}{a}$ 3) $\frac{a}{b}$ 4) $\frac{na}{b}$

42. $\lim_{h \rightarrow 0} \frac{(2+h)\cos(2+h) - 2\cos 2}{h} =$

- 1) $\cos 2 - 2\sin 2$ 2) $\cos 2 + 2\sin 2$
 3) $\sin 2 - 2\cos 2$ 4) $\sin 2 + 2\cos 2$

43. Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$,

then the value of $\lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{h^3 + 3h}$ is

- 1) $-\frac{50}{3}$ 2) $\frac{22}{3}$ 3) 13 4) $\frac{53}{3}$

PROBLEMS USING EXPANSIONS

44. $\lim_{x \rightarrow 0} \frac{\log(1+x) - x}{x^2} =$

- 1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $-\frac{1}{3}$

PROBLEMS USING SANDWICH THEOREM

45. $\lim_{x \rightarrow 0} x^2 \sin \frac{\pi}{x} =$

- 1) 1 2) 0 3) does not exist 4) ∞

46. $\lim_{x \rightarrow \infty} \frac{x^2(2 + \sin^2 x)}{x + 100} =$

- 1) 0 2) 1 3) ∞ 4) does not exist

47. $\lim_{x \rightarrow 0} \frac{x}{a} \left[\frac{b}{x} \right] (a \neq 0)$ [where $[]$ denotes the G.I.F.) is equal to

- 1) a 2) b 3) $\frac{b}{a}$ 4) $1 - \frac{b}{a}$

LEVEL-I (H.W)-KEY

- 01) 1 02) 1 03) 3 04) 4 05) 2 06) 1
07) 3 08) 3 09) 2 10) 1 11) 2 12) 1
13) 1 14) 3 15) 4 16) 4 17) 3 18) 2
19) 4 20) 4 21) 1 22) 2 23) 2 24) 1
25) 1 26) 1 27) 2 28) 4 29) 2 30) 2
31) 2 32) 3 33) 3 34) 3 35) 4 36) 3
37) 1 38) 1 39) 2 40) 1 41) 2 42) 1
43) 4 44) 2 45) 2 46) 3 47) 3

LEVEL-I (H.W)-HINTS

1. Using

2. Using L-Hospital rule

3. Using L-Hospital rule

4. Using L-Hospital rule

5. Using L-Hospital rule

6. On rationalizing given limit is

7. On applying limit

8.

9.

10. Use L-H rule

11. Given limit is

12. Given limit is

13. Given limit is

14. Given limit is

15. Given limit

16. Given limit is

17.

18. Use LH Rule

19. Use L-Hospital rule

20. Use L-Hospital rule

21. Use L-Hospital rule

22. Use L-Hospital rule

23.

24. Use L-Hospital rule

25. Rationalising

26. Divide with x in Nr and Dr

27.

28. Divide with

29. Given limit

30. Given limit

31. Given limit =

32. Divide with

33. Given limit is

34. Given limit

35.

$$36. \lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x^2 - 1} \right)^{x^2} = e^{\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x^2 - 1} - 1 \right)} = e^2$$

$$37. \lim_{x \rightarrow \infty} \left(\frac{3x - 4}{3x + 2} \right)^{\frac{x+1}{3}} = \lim_{x \rightarrow \infty} e^{\frac{x+1}{3} \left(\frac{3x-4}{3x+2} - 1 \right)} = e^{\frac{-2}{3}}$$

$$38. \lim_{x \rightarrow 1} x^{\left(\frac{1}{1-x^2} \right)} = e^{\lim_{x \rightarrow 1} \left(\frac{1}{1-x^2} \right) (x-1)} = e^{\frac{-1}{2}}$$

39. Use factorisation in power.

40.

41. By L- Hospital rule

Given limit =

42. By L = Hospital

43. Use LH Rule

44. use log (1+x) expansion

45.

46. Use Sandwich theorem (or) divide by

47.



LEVEL-II - (C.W)

EVALUATION OF ALGEBRA OF LIMITS

- $$\lim_{x \rightarrow 1} \frac{\left(\sum_{K=1}^{200} x^K\right) - 200}{x-1} =$$

1) 5050 2) 1000 3) 2010 4) 20100
- $$\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} =$$

1) $\frac{2}{\sqrt{3}}$ 2) $-\frac{1}{\sqrt{3}}$ 3) $\frac{2}{3\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$
- Let α and β be the roots of $ax^2+bx+c=0$,
 then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2} =$

1) $\frac{a^2(\alpha - \beta)^2}{2}$ 2) $\frac{a^2}{2(\alpha - \beta)^2}$
 3) $\frac{a^2}{(\alpha - \beta)^2}$ 4) $-\frac{a^2}{2(\alpha - \beta)^2}$
- If $f(x) = \begin{cases} \sin x, & x \neq n\pi, n \in \mathbb{Z} \\ 2, & \text{otherwise} \end{cases}$ and

$$g(x) = \begin{cases} x^2 + 1, & x \neq 0, 2 \\ 4, & x = 0 \\ 5, & x = 2 \end{cases}$$

then $\lim_{x \rightarrow 0} g[f(x)] =$

1) 2 2) 4 3) 5 4) 1
- $$\lim_{y \rightarrow x} \frac{y^y - x^x}{y - x} =$$

1) xy^{x-1} 2) $x^x(1 + \log x)$ 3) $y \cdot x^{y-1}$ 4) xy
- If $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 1}{x + 1} - ax - b \right) = 2$, then

1) $a = 1$ and $b = -3$ 2) $a = 1$ and $b = 2$
 3) $a = 0$ and $b = -1$ 4) $a = 2$ and $b = 1$

EVALUATION OF LEFT & RIGHT HAND LIMITS

7. Let $\{x\}$ denote the fractional part of x .

Then $\lim_{x \rightarrow 0} \frac{\{x\}}{\tan\{x\}}$ is equal to

- 1) -1 2) 0 3) 1 4) Does not exist

8. $\lim_{x \rightarrow 4^+} \frac{x^2 - 7x + 12}{x - [x]} =$ [where $[]$ denotes G.I.F.]
 1) 1 2) 0 3) 2 4) does not exist

9. If $f: R \rightarrow R$ defined by

$$f(x) = \begin{cases} \frac{x-2}{x^2-3x+2} & \text{if } x \in R - \{1, 2\} \\ 2 & \text{if } x = 1 \\ 1 & \text{if } x = 2 \end{cases}$$

then $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} =$

- 1) 0 2) -1 3) 1 4) -1/2

10. $\lim_{x \rightarrow 0} \frac{\sqrt{\frac{1}{2}(1 - \cos x)}}{x} =$

- 1) 1 2) -1 3) 0 4) does not exist

EVALUATION OF TRIGONOMETRIC LIMITS

11. If $y = \frac{1}{2} \sin^{-1} \left(\frac{2xy}{x^2 + y^2} \right)$ and $y < x$ then

$\lim_{y \rightarrow 0} x =$

- 1) -1 2) 0 3) 1 4) ∞

12. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos x - \sin x}{(4x - \pi)^2} =$

- 1) $\frac{1}{16\sqrt{2}}$ 2) $\frac{1}{32\sqrt{2}}$ 3) $\frac{1}{16}$ 4) $\frac{1}{8}$

13. $\lim_{x \rightarrow 0} \frac{\sec x - 1}{x^2 (\sec x + 1)^2} =$

- 1) 1/8 2) 1/4 3) 2 4) 0

14. $\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{\sinh^2 x} \right) =$

- 1) $\frac{2}{3}$ 2) 0 3) $\frac{1}{3}$ 4) $-\frac{2}{3}$

EVALUATION OF EXPONENTIAL & LOGARITHMIC LIMITS

15. $\lim_{x \rightarrow \infty} x[\log(x+1) - \log x] =$

- 1) e^2 2) e 3) 1 4) $1/e$

16. $\lim_{x \rightarrow 0} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}} =$
 1) 0 2) $8\sqrt{2}$ 3) $8(\log 3)^2$ 4) 1

17. $\lim_{x \rightarrow \infty} x(a^{\frac{1}{x}} - b^{\frac{1}{x}}) =$
 1) 1 2) $\log_e a/b$ 3) $\log_e(ab)$ 4) 0

LIMITS OF THE FORM TENDS TO INFINITE

18. The value of

$\lim_{n \rightarrow \infty} \frac{1.2 + 2.3 + 3.4 + \dots + n.(n+1)}{n^3}$ is
 1) 1 2) -1 3) $1/3$ 4) $-1/3$

19. If $|x| < 1$, then

$\lim_{n \rightarrow \infty} (1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n}) =$
 1) $\frac{1}{x}$ 2) $\frac{1}{1+x}$ 3) $\frac{1}{1-x}$ 4) $\frac{1}{x-1}$

20. $\lim_{n \rightarrow \infty} \frac{2.3^{n+1} - 3.5^{n+1}}{2.3^n + 3.5^n} =$
 1) 5 2) $1/5$ 3) -5 4) 0

21. $\lim_{n \rightarrow \infty} \frac{1}{n^4} [1^2 + (1^2 + 2^2) + \dots + (1^2 + 2^2 + \dots + n^2)] =$
 1) $1/6$ 2) $1/16$ 3) $1/12$ 4) 0

22. $\lim_{n \rightarrow \infty} \frac{{}^n P_n}{{}^{n+1} P_{n+1} - {}^n P_n} =$
 1) 2 2) -1 3) 0 4) ∞

23. If $a > 0$, $\lim_{x \rightarrow \infty} \frac{[ax + b]}{x}$ is [where [.] denotes G.I.F]
 1) 0 2) 1 3) a 4) b

LIMIT OF THE FORM $0^0, 1^\infty, \infty^0$

24. $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x + \dots + n^x}{n} \right)^{\frac{1}{x}} =$
 1) $(n!)^n$ 2) $(n!)^{1/n}$ 3) $n!$ 4) $\ln n!$

25. $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{\sin x}} =$
 1) 1 2) -1 3) 0 4) 3

26. $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}} =$
 1) e 2) e^2 3) e^3 4) $1/e$

27. If $\lim_{x \rightarrow 0} [1 + x \ln(1 + b^2)]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$
 and $\theta \in (-\pi, \pi)$ then the value of θ is

1) $\pm \frac{\pi}{6}$ 2) $\pm \frac{\pi}{3}$ 3) $\pm \frac{\pi}{8}$ 4) $\pm \frac{\pi}{2}$

28. $\lim_{x \rightarrow 1} (\log_3 3x)^{\log_3 3} =$

1) e^{-1} 2) e 3) -1 4) 1

29. If p and q are the roots of the quadratic equation $ax^2 + bx + c = 0$ then

$\lim_{x \rightarrow p} (1 + ax^2 + bx + c)^{\frac{1}{x-p}} =$

1) $a(p-q)$ 2) $\log[a(p-q)]$
 3) $e^{a(p-q)}$ 4) $e^{a(q-p)}$

30. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x =$
 1) e^4 2) e^3 3) e^2 4) 2^4

31. $\lim_{x \rightarrow \infty} \left\{ \left(\frac{x}{x+1} \right)^a + \sin \frac{1}{x} \right\}^x$ is equal to
 1) e^{a-1} 2) e^{1-a} 3) e 4) 0

32. $\lim_{x \rightarrow 0} \left\{ \tan \left(\frac{\pi}{4} + x \right) \right\}^{\frac{1}{x}} =$
 1) e 2) e^2 3) e^{-1} 4) e^{-2}

33. $\lim_{x \rightarrow \infty} \left(1 + \frac{\lambda}{x} + \frac{\mu}{x^2} \right)^{2x} = e^4$ then $\lambda = (\mu \in \mathbb{R})$
 1) 1 2) 2 3) 3 4) 4

L.H.RULE

34. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is such that $f(1) = 3$ and $f'(1) = 6$. Then

$\lim_{x \rightarrow 0} \left[\frac{f(1+x)}{f(1)} \right]^{\frac{1}{x}} =$
 1) 1 2) e^2 3) $e^{1/2}$ 4) e^3

PROBLEMS USING EXPANSIONS

35. The integer n for which

$\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number is

- 1) 1 2) 2 3) 3 4) 4

36. The values of a, b , and c such that

$$\lim_{x \rightarrow 0} \frac{ax e^x - b \log(1+x) + cx e^{-x}}{x^2 \sin x} = 2$$

- 1) $a = 3, b = 12, c = 9$
 2) $a = 1, b = 2, c = 4$
 3) $a = 2, b = 10, c = 8$
 4) $a = 3, b = -12, c = -9$

37. $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - \sin x}{x^3}$

- 1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{5}$

38. $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e\left(1 - \frac{x}{2}\right)}{(1 - \cos x)}$

- 1) $\frac{1}{2}e$ 2) $\frac{1}{4}e$ 3) $\frac{11}{12}e$ 4) $\frac{1}{12}e$

PROBLEMS USING SANDWICH THEOREM

39. If $[x]$ denotes the greatest integer less than

or equal to x then $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2} =$

- 1) $x/2$ 2) $x/3$ 3) x 4) 0

40. $\lim_{n \rightarrow \infty} \frac{\{x\} + \{2x\} + \dots + \{nx\}}{n^2}$

- 1) $\frac{1}{2}$ 2) 0 3) -1 4) 2

41. $\lim_{x \rightarrow 0} \left[\frac{\sin|x|}{|x|} \right]$ Where $[.]$ denotes the greatest integer function.

- 1) 0 2) 1 3) -1 4) does not exist

42. Let $f: R \rightarrow R$ be a positive increasing

function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$, then

$$\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = \quad \text{[AIEEE 2010]}$$

- 1) $\frac{2}{3}$ 2) $\frac{3}{2}$ 3) 3 4) 1

LEVEL-II (C.W)-KEY

- 1) 4 2) 3 3) 1 4) 4 5) 2 6) 1
 7) 4 8) 1 9) 2 10) 4 11) 3 12) 1
 13) 1 14) 1 15) 3 16) 2 17) 2 18) 3
 19) 3 20) 3 21) 3 22) 3 23) 3 24) 2
 25) 1 26) 4 27) 4 28) 2 29) 3 30) 1
 31) 2 32) 2 33) 2 34) 2 35) 3 36) 1
 37) 2 38) 3 39) 1 40) 2 41) 1 42) 4

LEVEL-II (C.W)-HINTS

1. Use L-Hospital rule ☐ Given Limit

2. Using L-Hospital rule given limit is

3.
given limit is

4. Find

5. Using L-Hospital rule,

6.

7. L.H.L

R.H.L.

☐ L.H.L ☐ R.H.L.

8. As

9.

10.

does not exist

11.

12. Use L-Hospital rule

$$13. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \cdot \left[\frac{1}{\cos x} \cdot \frac{1}{(\sec x + 1)^2} \right] = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

14. Use L-Hospital rule

15.

16.

$$17. \lim_{x \rightarrow \infty} \frac{a^{\frac{1}{x}} - b^{\frac{1}{x}}}{\frac{1}{x}} = \log \left(\frac{a}{b} \right)$$

18.

19.

20.

21.

22.

23.

24.

25.

$$26. \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}} = e^{\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} - 1 \right) \frac{\sin x}{x - \sin x}} = 1/e$$

27.

28.

29.

$$30. \lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x = 1^\infty \Rightarrow e^{\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} - 1 \right) x} = e^4$$

LIMITS

31. Given limit

where

32.

33.

34.

35.

is non-zero if $n = 3$

36.

coefficient of x & and coefficient of

37.

38.

39. By using Sandwich theorem

JEE-MAIN-JR-MATHS VOL-IV

40.

41. If

then

42. By using Sandwich theorem



LEVEL-II - (H.W)

EVALUATION OF ALGEBRA OF LIMITS

1. $\lim_{x \rightarrow 0} \frac{(1+x)^3 - 1 - 3x}{(1+x)^2 - 1 - 2x} =$

- 1) 2 2) 3 3) 5 4) -2

2. If $f(x) = \frac{4-7x}{7x+4}$, $\lim_{x \rightarrow 0} f(x) = l$ and

$\lim_{x \rightarrow \infty} f(x) = m$ the quadratic equation

having roots as $\frac{1}{l}$ and $\frac{1}{m}$ is

- 1) $x^2 - 1 = 0$ 2) $x^2 + 1 = 0$
3) $1/2$ 4) $x^3 - 1 = 0$

3. $\lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3\sqrt{2x-3}}{\sqrt[3]{x+6} - 2\sqrt[3]{3x-5}} =$

- 1) $\frac{34}{23}$ 2) $\frac{23}{17}$ 3) $\frac{7}{23}$ 4) $\frac{23}{7}$

4. $\lim_{x \rightarrow 1} \left[\frac{x^3 + 2x^2 + x + 1}{x^2 + 2x + 3} \right]^{\frac{1 - \cos(x-1)}{(x-1)^2}} =$
- 1) e 2) $e^{1/2}$ 3) 1 4) $\left(\frac{5}{6}\right)^{1/2}$
5. $\lim_{x \rightarrow 0} \frac{x\sqrt{y^2 - (y-x)^2}}{\left\{ \sqrt{(8xy - 4x^2)} + \sqrt{8xy} \right\}^3} =$
- 1) $\frac{1}{4y}$ 2) $\frac{1}{2}$ 3) $\frac{1}{2\sqrt{2}}$ 4) $\frac{1}{128y}$
6. $\lim_{x \rightarrow \infty} \frac{x^5}{5^x} =$ _____
- 1) 0 2) 1 3) ∞ 4) does not exist
7. $\lim_{x \rightarrow 8} \frac{\sqrt{1 + \sqrt{1 + x}} - 2}{x - 8} =$ [EAMCET - 2011]
- 1) $\frac{3}{2}$ 2) $\frac{1}{4}$ 3) $\frac{1}{24}$ 4) $\frac{1}{5}$

EVALUATION OF LEFT & RIGHT HAND LIMITS

8. $\lim_{x \rightarrow 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$ [AIEEE 2011]
- 1) equals $\sqrt{2}$ 2) equals $-\sqrt{2}$
- 3) equals $\frac{1}{\sqrt{2}}$ 4) does not exist
9. If $l_1 = \lim_{x \rightarrow 2^+} (x + [x]), l_2 = \lim_{x \rightarrow 2^-} (2x - [x])$ and $l_3 = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}}$ then [where [] denotes G.I.F.]
- 1) $l_1 < l_2 < l_3$ 2) $l_2 < l_3 < l_1$
- 3) $l_3 < l_2 < l_1$ 4) $l_1 < l_3 < l_2$

EVALUATION OF TRIGONOMETRIC LIMITS

10. If $a = \min\{x^2 + 4x + 5, x \in R\}$ and $b = \lim_{\theta \rightarrow 0} \frac{1 - \cos 2\theta}{\theta^2}$ then the value of

$$\sum_{r=0}^n a^r b^{n-r} =$$

- 1) $\frac{2^{n+1} - 1}{4 \cdot 2^n}$ 2) $2^{n+1} - 1$
- 3) $\frac{2^{n+1} - 1}{3 \cdot 2^n}$ 4) $2^n - 1$
11. $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^2} =$ [EAMCET - 2010]
- 1) 0 2) 1 3) $\frac{1}{2}$ 4) $-\frac{1}{2}$
12. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x} =$ [JEE M 2013]
- 1) $-\frac{1}{4}$ 2) $\frac{1}{2}$ 3) 1 4) 2
13. $\lim_{x \rightarrow 0} \frac{\tan^3 x - \sin^3 x}{x^5} =$ [EAMCET - 2013]
- 1) $\frac{5}{2}$ 2) $\frac{3}{2}$ 3) $\frac{3}{5}$ 4) $\frac{2}{5}$
14. $\lim_{x \rightarrow 0} \frac{\sin 2x + 2\sin^2 x - 2\sin x}{\cos x - \cos^2 x} =$
- 1) 0 2) 1 3) $\frac{2}{3}$ 4) 4
15. $\lim_{x \rightarrow 0} \frac{3 \sin x - \sin 3x}{x \cdot \tan^2 2x} =$
- 1) 0 2) 1 3) 2 4) 4
16. $\lim_{x \rightarrow 0} \frac{8}{x^8} \left(1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cdot \cos \frac{x^2}{4} \right) =$
- 1) $\frac{1}{16}$ 2) $\frac{1}{15}$ 3) $\frac{1}{32}$ 4) 1
17. Arrange the following limits in the ascending order.
- 1) $\lim_{x \rightarrow 0} \frac{\tan^4 x - \sin^4 x}{x^6}$ 2) $\lim_{x \rightarrow 0} \frac{\tan^8 x - \sin^8 x}{x^5 \tan x^5}$
- 3) $\lim_{x \rightarrow 0} \frac{\tan^3 x - \sin^3 x}{x \sin^4 x}$ 4) $\lim_{x \rightarrow 0} \frac{\tan^5 x - \sin^5 x}{x^2 \cdot \sinh^3 x \cdot \tan^2 x}$
- 1) 1, 2, 3, 4 2) 3, 1, 4, 2
- 3) 1, 2, 4, 3 4) 2, 1, 3, 4

18. The value of θ , is

$$\lim_{\theta \rightarrow 0} \frac{\cos^2 \left\{ 1 - \cos^2 \left(1 - \cos^2 \dots \left(\cos^2 \{ 1 - \cos^2 \theta \} \right) \right) \right\}}{\sin \left(\frac{\pi(\sqrt{\theta+4}-2)}{\theta} \right)}$$

- 1) $\frac{\sqrt{2}}{4}$ 2) $\sqrt{2}$ 3) 1 4) 2

19. $\lim_{x \rightarrow \pi} \frac{\sqrt{2+\cos x}-1}{(\pi-x)^2} =$ _____

- 1) 0 2) $\frac{1}{4}$ 3) $\frac{1}{2}$ 4) 2

20. If $\lim_{x \rightarrow 0} \frac{\{(a-n)x - \tan x\} \sin nx}{x^2} = 0$, where n is a non-zero real number, then 'a' =

- 1) 0 2) $\frac{n+1}{n}$ 3) n 4) $n + \frac{1}{n}$

21. $\lim_{h \rightarrow 0} 2 \left[\frac{\sqrt{3} \sin \left(\frac{\pi}{6} + h \right) - \cos \left(\frac{\pi}{6} + h \right)}{\sqrt{3} \cdot h (\sqrt{3} \cosh - \sinh)} \right] =$

- 1) $-\frac{2}{\sqrt{3}}$ 2) $-\frac{4}{3}$ 3) $\frac{2}{3}$ 4) $\frac{4}{3}$

EVALUATION OF EXPONENTIAL & LOGARITHMIC LIMITS

22. The value of $\lim_{x \rightarrow a} \frac{\log(x-a)}{\log(e^x - e^a)}$ is

- 1) 1 2) -1 3) 0 4) 2

23. Arrange the following limits in the ascending order

1) $\lim_{x \rightarrow \infty} \left(\frac{1+x}{2+x} \right)^{x+2}$ 2) $\lim_{x \rightarrow 0} (1+2x)^{3/x}$

3) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{2\theta}$ 4) $\lim_{x \rightarrow 0} \frac{\log_e(1+x)}{x}$

- 1) 1, 2, 3, 4 2) 1, 3, 4, 2
3) 1, 4, 3, 2 4) 3, 4, 1, 2

24. $\lim_{x \rightarrow 0} \frac{(4^x - 1)^3}{\sin \left(\frac{x}{4} \right) \log_e \left(1 + \frac{x^2}{3} \right)} =$

1) $(\log_e 4)^3$ 2) $\log_e 4$

3) $12(\log_e 4)^3$ 4) $5(\log_e 4)^3$

LIMITS OF THE FORM TENDS TO INFINITE

25. $\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} =$

- 1) 1 2) -1 3) 0 4) does not exist

26. $\lim_{n \rightarrow \infty} \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{3n^4 + 5n^3 + 6} =$

- 1) 1/3 2) 1/5 3) 1/6 4) 1/12

27. $\lim_{x \rightarrow \infty} \frac{3\sqrt{x} + 5\sin^2 x - 10 \cdot \log x}{5\sqrt{x} + 7\cos^2 x + 100 \cdot \log x} =$

- 1) $\frac{3}{5}$ 2) $\frac{5}{3}$ 3) 15 4) $\frac{1}{15}$

28. $\lim_{n \rightarrow \infty} \frac{(\sqrt{n^2 + 1} + n)^2}{\sqrt[3]{n^6 + 1}} =$

- 1) 1 2) 1/2 3) 1/3 4) 4

29. $\lim_{x \rightarrow \infty} \frac{(2+x)^{20}(4+x)^3}{(2-x)^{23}} =$

- 1) -1 2) 1 3) 6 4) 2

30. $\lim_{n \rightarrow \infty} \cos \left(\pi \sqrt{n^2 + n} \right)$ is equal to

- 1) 0 2) 1 3) 2 4) does not exist

31. $\lim_{n \rightarrow \infty} \left(\frac{1}{5} \right)^{\log_{\sqrt{5}} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \text{to } n \text{ terms} \right)}$ equals

- 1) 2 2) 4 3) 8 4) 0

32. $\lim_{n \rightarrow \infty} \frac{1^4 + 2^4 + 3^4 + \dots + n^4}{n^5}$

$-\lim_{n \rightarrow \infty} \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^5}$ is

- 1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

33. If $a > 1$, then the value of $\lim_{x \rightarrow \infty} \frac{a^{\sqrt{x}} - a^{1/\sqrt{x}}}{a^{\sqrt{x}} + a^{1/\sqrt{x}}}$ is

- 1) 1 2) 0 3) 2 4) 3

LIMIT OF THE FORM $0^0, 1^\infty, \infty^0$

34. for $x > 0$; $\lim_{x \rightarrow 0} \left((\sin x)^{\frac{1}{x}} + \left(\frac{1}{x} \right)^{\sin x} \right) =$

- 1) 0 2) -1 3) 1 4) 2

35. $\lim_{x \rightarrow 0} \left(\frac{1 + \tan x}{1 + \sin x} \right)^{\csc x}$ is equal to

- 1) $\frac{1}{e}$ 2) e 3) e^2 4) 1

36. $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4} =$ [EAMCET - 2012]

- 1) e^4 2) e^6 3) e^5 4) e

37. $\lim_{x \rightarrow \infty} \left(\frac{x+5}{x+2} \right)^{x+3} =$ [EAMCET - 2009]

- 1) e 2) e^2 3) e^3 4) e^5

L.H.RULE

38. If $f'(0) = 3$, then

$$\lim_{x \rightarrow 0} \frac{x^2}{f(x^2) - 6f(4x^2) + 5f(7x^2)} =$$

- 1) $\frac{1}{36}$ 2) $-\frac{1}{36}$ 3) $\frac{1}{34}$ 4) $\frac{1}{106}$

PROBLEMS USING EXPANSIONS

39. $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x} =$

- 1) 1 2) $e/2$ 3) $-e/2$ 4) $2/e$

PROBLEMS USING SANDWICH THEOREM

40. If $[x]$ denotes the greatest integer less than or equal to x then

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \left\{ [1^2 x] + [2^2 x] + [3^2 x] + \dots + [n^2 x] \right\} =$$

- 1) $\frac{x}{2}$ 2) $\frac{x}{3}$ 3) $\frac{x}{6}$ 4) 0

LEVEL-II (H.W)-KEY

- 1) 2 2) 1 3) 1 4) 4 5) 4 6) 1
 7) 3 8) 4 9) 3 10) 2 11) 1 12) 4
 13) 2 14) 4 15) 2 16) 3 17) 2 18) 2
 19) 2 20) 4 21) 4 22) 1 23) 2 24) 3
 25) 4 26) 4 27) 1 28) 4 29) 1 30) 1
 31) 2 32) 1 33) 1 34) 3 35) 4 36) 3
 37) 3 38) 1 39) 3 40) 2

LEVEL-II (H.W)-HINTS

1. Using L-Hospital rule given limit is

2. ,

3. Using L-Hospital rule

4.

5. Take common higher power of x in both numerator and denominator

6.

where

7. Use L-Hospital's rule

8.

9. Definition of and L hospital rule

10.

11. Take common and simplify

12.

13.

14. Given limit is

15. $\lim_{x \rightarrow 0} \frac{4 \sin^3 x}{x \tan^2 2x} = \frac{4}{4} = 1$

16. Given limit is

17.

18. Given limit

19.

= =

=

=

20. ,

21. Using L.H Rule

22.

23. Using standard formulae

24. Divide with and simplify

25. L.H.L=

R.H.L=

26. Given limit

27. Divide with

28. $\frac{(1+1)^3}{\sqrt[3]{1+0}} = 4$

29. Divide with and applying limit

30.

31.

32.
33.
34. and form
35. form
36.
37.
38. By L-Hospital rule
39.
40. Sandwich theorem



LEVEL-III - (C.W)

1. $\lim_{x \rightarrow 0} \frac{e^{x^3} - 1 - x^3}{\sin^6(2x)} =$
 1) $\frac{1}{128}$ 2) $\frac{2}{127}$ 3) $\frac{1}{126}$ 4) $\frac{1}{125}$
2. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - (\sin x)^{\sin x}}{\cos^2 x} =$
 1) 2 2) 1 3) $1/2$ 4) $1/4$
3. If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$ then $\lim_{x \rightarrow 0} \frac{f'(x)}{x} =$
 1) 1 2) -1 3) 2 4) -2
4. $\lim_{x \rightarrow 0} \left(\frac{\log \sec \left(\frac{x}{2} \right) \cos x}{\log \sec x \cos \left(\frac{x}{2} \right)} \right) =$
 1) 14 2) 15 3) 16 4) 17

5. $\lim_{x \rightarrow 0} \left[\frac{100 \tan x \cdot \sin x}{x^2} \right]$ where $[.]$

represents greatest integer function is

- 1) 99 2) 100 3) 0 4) 98

6. If $\{x\}$ denotes fractional part of x , then

$\lim_{x \rightarrow 1} \frac{x \sin \{x\}}{x - 1} =$

- 1) 0 2) -1 3) 1 4) does not exist

7. The graph of the function $y = f(x)$ has a unique tangent at the point $(e^a, 0)$ through which the graph passes then

$\lim_{x \rightarrow e^a} \frac{\log_e \{1 + 7f(x)\} - \sin f(x)}{3f(x)}$ is

- 1) 1 2) 2 3) 0 4) -1

8. If $[.]$ denotes the greatest integer

function, then $\lim_{x \rightarrow \frac{\pi}{2}} \left[\frac{x - \frac{\pi}{2}}{\cos x} \right] =$

- 1) 1 2) -1 3) 2 4) -2

9. If $[.]$ denotes the greatest integer

function then $\lim_{x \rightarrow 0} \left[\frac{x^2}{\tan x \cdot \sin x} \right] =$

- 1) 0 2) 1 3) -1 4) does not exist

10. $\lim_{n \rightarrow \infty} \frac{1}{n^4} \sum_{r=1}^n r(r+2)(r+4) =$

- 1) $\frac{3}{4}$ 2) 0 3) $\frac{1}{8}$ 4) $\frac{1}{4}$

11. $\lim_{n \rightarrow \infty} \left[\frac{7}{10} + \frac{29}{10^2} + \frac{133}{10^3} + \dots + \frac{5^n + 2^n}{10^n} \right] =$

- 1) $3/4$ 2) 2 3) $5/4$ 4) $1/2$

12. Suppose $f(n+1) = \frac{1}{2} \left\{ f(n) + \frac{9}{f(n)} \right\}, n \in N$. If

$f(n) > 0, \forall n \in N$, then $\lim_{n \rightarrow \infty} (f(n)) =$

- 1) 3^{-1} 2) -3^{-1} 3) 3 4) -3

13. $\lim_{x \rightarrow 0} \left[1^{1/\sin^2 x} + 2^{1/\sin^2 x} + \dots + n^{1/\sin^2 x} \right]^{\sin^2 x} =$
 1) ∞ 2) 0 3) $\frac{n+1}{2}$ 4) n

14. The value of

$\lim_{x \rightarrow 0} \left(\frac{e^{\lfloor \log(2^x - 1) \rfloor} - (2^x - 1)^x \sin x}{e^{x \log x}} \right)^{1/x}$ is equal to

- 1) e 2) $\frac{1}{e} \log 2$ 3) $e \log 2$ 4) $\log 2$

15. $\lim_{n \rightarrow \infty} (n^4 + n^3 + A_1 n^2 + A_2 n + A_3)^{1/2} - (n^4 + n^3 + B_1 n^2 + B_2 n + B_3)^{1/2}$ equals

- 1) $\frac{A_1 - B_1}{2}$ 2) $\frac{A_1 + B_3}{2}$
 3) $\frac{B_3 - A_1}{2}$ 4) $\frac{A_1 - B_3}{2}$

16. $\lim_{n \rightarrow \infty} \frac{1 \cdot n^2 + 2(n-1)^2 + 3(n-2)^2 + \dots + n \cdot 1^2}{1^3 + 2^3 + \dots + n^3} =$

- 1) $\frac{8}{3}$ 2) $\frac{4}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$

17. $\lim_{n \rightarrow \infty} \frac{1 - 2 + 3 - 4 + 5 - 6 + \dots - 2n}{\sqrt{n^2 + 1} + \sqrt{4n^2 - 1}} =$

- 1) $\frac{1}{3}$ 2) $-\frac{1}{3}$ 3) $-\frac{1}{5}$ 4) $\frac{1}{5}$

18. If $\lim_{x \rightarrow 0} (x^{-3} \sin 3x + ax^{-2} + b)$ exists and is equal to zero, then the value of $a + 2b =$
 1) 3 2) 4 3) 0 4) 6

19. The graph of $y = f(x)$ has unique tangent at the point $(a, 0)$ through which the graph passes. Then

$\lim_{x \rightarrow a} \frac{\log[1 + 6f(x)]}{3f(x)} =$

- 1) 0 2) 1 3) 2 4) ∞

20. The value of

$\lim_{x \rightarrow 0} \left(1 - \frac{1}{2^x} \right) \left(\frac{1}{\sqrt{\tan x + 4} - 2} \right)$ is

- 1) $\log 16$ 2) does not exist
 3) $3 \log 2$ 4) $6 \log 2$

21. The value of

$\lim_{x \rightarrow 0} \left\{ \left\lfloor \frac{100x}{\sin x} \right\rfloor + \left\lfloor \frac{99 \sin x}{x} \right\rfloor \right\}$, where $[.]$

represents the greatest integer function, is

- 1) 199 2) 198 3) 0 4) 1

22. $\lim_{x \rightarrow 0^-} \frac{[x] + [x^2] + [x^3] + \dots + [x^{2n+1}] + n + 1}{1 + [x^2] + [x] + 2x}$

$n \in N$ is equal to

- 1) $n+1$ 2) n 3) 1 4) 0

23. If $[.]$ denotes the greatest integer function, then

$\lim_{x \rightarrow 0} \frac{\tan(\lfloor -2\pi^2 \rfloor x^2) - x^2 \tan(\lfloor -2\pi^2 \rfloor)}{\sin^2 x} =$

- 1) $-20 + \tan 20$ 2) $20 + \tan 20$
 3) 20 4) $\tan 20$

24. $\lim_{n \rightarrow \infty} \frac{2^3 - 1^3}{2^3 + 1^3} \cdot \frac{3^3 - 1^3}{3^3 + 1^3} \dots \frac{n^3 - 1}{n^3 + 1} =$

- 1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) 1 4) $\frac{3}{2}$

25. $\lim_{n \rightarrow \infty} {}^n C_r \left(\frac{m}{n} \right)^r \left(1 - \frac{m}{n} \right)^{n-r} =$

- 1) $e^{-m} m^r$ 2) $\frac{m^r}{r!}$ 3) $\frac{m^r e^{-m}}{r!}$ 4) $\frac{e^{-r} r^m}{m!}$

26. The value of $\lim_{x \rightarrow 0} \left\{ \sin^2 \left(\frac{\pi}{2 - ax} \right) \right\}^{\sec^2 \frac{\pi}{2 - bx}} =$

- 1) $e^{-a/b}$ 2) e^{-a^2/b^2} 3) $a^{2a/b}$ 4) $e^{4a/b}$

27. $\lim_{x \rightarrow \infty} \left[\frac{1^2}{1-x^3} + \frac{3}{1+x^2} + \frac{5^2}{1-x^3} + \frac{7}{1+x^2} + \dots \right] =$

- 1) $-\frac{5}{6}$ 2) $-\frac{10}{3}$ 3) $\frac{5}{6}$ 4) $\frac{10}{3}$

28. Evaluate $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{a_1} \right) \left(1 + \frac{1}{a_2} \right) \dots \left(1 + \frac{1}{a_n} \right)$

where $a_1 = 1$ and $a_n = n(1 + a_{n-1}) \forall n \geq 2$

- 1) $\frac{1}{e}$ 2) $\frac{1}{e^2}$ 3) e 4) 1

29. $\lim_{n \rightarrow \infty} n^{-n^2} \left\{ (n+1) \left(n + \frac{1}{2} \right) \left(n + \frac{1}{2^2} \right) \dots \left(n + \frac{1}{2^{n-1}} \right) \right\}^n =$

- 1) $\frac{1}{2e^2}$ 2) $\frac{1}{e}$ 3) e^2 4) $\frac{3}{2}e^2$

30. Evaluate

$\lim_{n \rightarrow \infty} \left\{ \frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \frac{x}{2^2} + \dots + \frac{1}{2^n} \tan \frac{x}{2^n} \right\}$

- 1) $x \tan \frac{x}{2}$ 2) $\frac{1}{x} \cot \frac{x}{2}$
3) $\frac{x - \cot x}{2}$ 4) $\frac{1}{x} - \cot x$

31. $\lim_{x \rightarrow -\pi} \frac{|x + \pi|}{\sin x} =$

- 1) 1 2) -1 3) π 4) does not exist

32. If $\ell(x)$ is least integer not less than x and $g(x)$ is the greatest integer not greater than x then

$\lim_{x \rightarrow e+\pi} (\ell(x) + g(x)) =$

- 1) 1 2) 9 3) 11 4) 13

33. If $0 < P < 1$ then $\lim_{n \rightarrow \infty} \frac{n^P \sin^2(n!)}{n+1} =$

- 1) 0 2) 1 3) ∞ 4) $\frac{4}{3}$

34. $\lim_{x \rightarrow a^+} \frac{\{x\} \sin(x-a)}{(x-a)^2} =$ where $\{x\}$

denotes fractional part of x and $a \in \mathbb{N}$

- 1) 0 2) 1 3) a 4) 5

35. $\lim_{x \rightarrow 0} \left\{ \left[\frac{a \sin x}{x} \right] + \left[\frac{b \tan x}{x} \right] \right\} =$ $a, b \in \mathbb{N}$,

[where $[]$ denotes G.I.F.]

- 1) $a+b$ 2) $a+b-1$ 3) 0 4) $\frac{a+b}{2}$

36. $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4} =$

- 1) $\frac{1}{6}$ 2) $\frac{1}{5}$ 3) $\frac{1}{4}$ 4) $\frac{1}{2}$

37. $\lim_{x \rightarrow 0} \frac{1 - \sin[\cos x]}{[x] - [\sin x]} =$ (where $[x]$ denotes

greatest integral part of x)

- 1) 0 2) 1 3) 2 4) ∞

38. $\lim_{x \rightarrow -1} \frac{1}{\sqrt{|x| - \{-x\}}} =$ (where $\{x\}$ denotes fractional part of x)

- 1) does not exist 2) 1 3) ∞ 4) $\frac{1}{2}$

39. $\lim_{x \rightarrow 0} \left[\frac{\sin(\operatorname{sgn}(x))}{(\operatorname{sgn}(x))} \right] =$

(where $[x]$ denotes integral part of x)

- 1) 0 2) 1 3) -1 4) does not exist

40. If $f(x) = |x-1| - [x]$ (where $[x]$ is greatest integer less than or equal to x) then.

- 1) $f(1^+) = -1; f(1^-) = 0$
2) $f(1^+) = 0 = f(1^-)$
3) $\lim_{x \rightarrow 1} f(x)$ exists
4) Cannot say anything

41. $\lim_{x \rightarrow -2^-} \frac{ae^{\frac{1}{|x+2|}} - 1}{2 - e^{\frac{1}{|x+2|}}} = \lim_{x \rightarrow -2^+} \sin\left(\frac{x^4 - 16}{x^5 + 32}\right)$ then a is

- 1) $\sin\left(\frac{3}{5}\right)$ 2) 2 3) $\sin\left(\frac{2}{5}\right)$ 4) $\sin\left(\frac{1}{5}\right)$

42. If $\{x\}$ denotes fractional part of x then

$\lim_{x \rightarrow 1} \frac{x \sin\{x\}}{x-1}$

- 1) 0 2) -1 3) 1 4) does not exist

43. $\lim_{n \rightarrow \infty} \sum_{x=1}^{20} \cos^{2n}(x-10) =$

- 1) 0 2) 1 3) 19 4) 20

44. If $f(x)$ is a real number in $[0, 1]$, then the value of the function

$f(x) = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} (1 + \cos^{2m}(n! \pi x))$ is given by

- 1) 2 or 1 according as x is a rational or irrational
2) 2 or 1 according as x is irrational or rational
3) 1 for all x 4) 2 for all x

45. $\lim_{x \rightarrow \infty} \sec^{-1} \left(\frac{x}{x+1} \right)$ is equal to

- 1) 0 2) π 3) $\frac{\pi}{2}$ 4) Does not exist

46. If $n \in N$ then $\lim_{x \rightarrow n} (-1)^{[x]}$, where

$[x]$ denotes the greatest integer less than or equal to x , is equal to

- 1) $(-1)^n$ 2) $(-1)^{n-1}$
3) 0 4) does not exist

47. $\lim_{x \rightarrow -1} \frac{\cos 2 - \cos 2x}{x^2 - |x|}$ is equal to

- 1) 2 2) $\sin 2$ 3) $2 \sin 2$ 4) 0

48. The value of $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^n} \right)^x, n > 0$ is

- 1) 1, if $n < 1$ 2) 1, if $n > 1$
3) e , if $n > 1$ 4) e , if $n < 1$

49. $\lim_{x \rightarrow 0} \left(1 + x \left(1 + \frac{f(x)}{kx^2} \right) \right)^{1/x} = e^3$ and

$f(4) = 64$ then K has value

- 1) 1 2) 2 3) 4 4) 5

LEVEL-III-KEY

- 1) 1 2) 3 3) 4 4) 3 5) 1 6) 4
7) 2 8) 4 9) 1 10) 4 11) 3 12) 3
13) 4 14) 2 15) 1 16) 4 17) 2 18) 4
19) 3 20) 1 21) 2 22) 4 23) 1 24) 2
25) 3 26) 2 27) 2 28) 3 29) 3 30) 4
31) 4 32) 3 33) 1 34) 2 35) 2 36) 1
37) 4 38) 1 39) 1 40) 1 41) 3 42) 4
43) 2 44) 1 45) 4 46) 4 47) 3 48) 2
49) 2

LEVEL-III-HINTS

1.

2. Use L hospital rule

3.

4. Use L hospital rule

5. We have in the

6.

7.

Given limit

8. Given limit

where

9. in the nbd of 0

10.

11.

12.

13. Given limit

Use L-Hospital rule

Again use L-Hospital rule

27.

28.

(☐ use ☐ & ☐ given)

proceeding in a similar manner weight

29.

30

31.

proceeding like this we get

= =

☐ L.H.L ☐ R.H.L
☐ limit does not exist

32. Let

,

=

use

33.

as &

34.

35. but close to

but close to

b

36.

37.

use
, ,

38. L.H.L =

R.H.L =

L.H.L R.H.L limit does not exist

39.

use

40. Find and and verify

41.

,

42.

43.

where
 where

Here for and in all other cases it is zero

44.

for is rational
 as for is irrational

45.

is not defined

46.

47.

48.

49.



LEVEL - IV

1. I :

II :

- 1) Only I is true
2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

2. I :

 even

integer where $[.]$ denotes the greatest integer function.

II :

then $a = 2, b = 2$

- 1) Only I is true
2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

3. Arrange the following limits in the descending order

a)

b)

c)

d)

- 1) b,c,d,a 2) d,a,c,b
3) d,c, b, a 4) b,c, a, d

4. Assertion(A) : If $|x| < 1$ then

Reason (R):

- 1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) R is true but A is false

5. Assertion(A) _____ :

Reason(R) :

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) R is true but A is false

6. List - I List - II

a) 1)

b) 2)

c) 3)

d) 4) 1

Then the correct match for List - I from List - II

	a	b	c	d		a	b	c	d
1)	1	4	3	2	2)	1	2	3	4
3)	2	1	4	3	4)	1	2	4	3

7. If m, n are +ve integers and are non-zero real numbers

 then

List - I List - II

a) If $m = n$ 1) $s =$

b) If $m < n$ 2) $s =$

c) If $m > n$ 3) $s = 0$

Then the correct match for List - I from

List - II

	a	b	c		a	b	c
1)	1	2	3	2)	1	3	2
3)	2	1	3	4)	1	3	3

8. Match the following :

List-I List-II

- A) 1) $1/2$
- B) 2) -1
- C) 3) does not exist
- D) 4) $(abc)^{1/3}$

A	B	C	D
1) 1	2	3	4
2) 4	3	3	1
3) 4	3	2	1
4) 4	3	1	2

9. I :

II :

- 1) Only I is true 2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

10. I : If $a > 0$ then , [Where

denotes greatest integral part of .

II :

[Where denotes greatest integral part of .

- 1) Only I is true
2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

11. I :

[Where denotes greatest integral part of .

II :

- 1) Only I is true 2) Only II is true
3) Both I and II are true
4) Neither I nor II is true

LEVEL-IV-KEY

- 1) 3 2) 2 3) 2 4) 1 5) 3 6) 4
7) 2 8) 3 9) 3 10) 3 11) 3

LEVEL-IV-HINTS

1. 1)

2)

2. Using step function definition

3. By simplifying

4. A is true, R is true

5. A is true, R is false

6. form

7. Divide with

Divide with

8. Using standard formula

9. 1) Divide with

2) Divide with x

10.

11. Standard formula